

Project 4 – Edge Computing Smart Home Appliance (Interrupts & Safety)

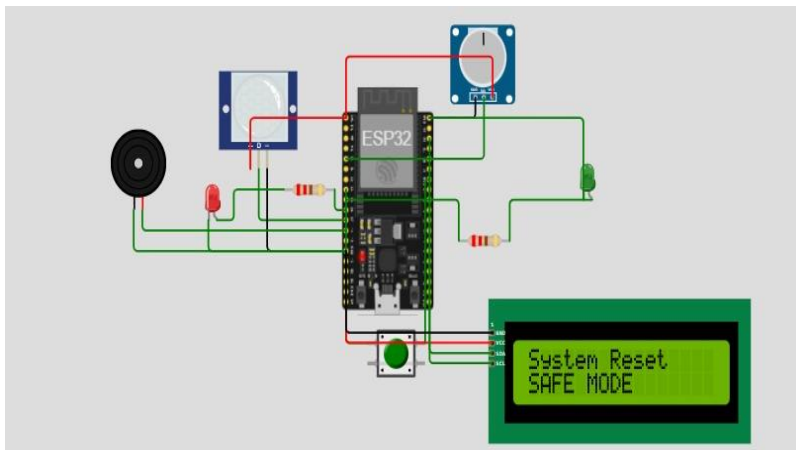
Objective:

The objective of this project is to design a smart home safety system using ESP32 that responds instantly to motion detection and emergency gas conditions using hardware interrupts and fail – safe logic.

Components used:

- ESP32
- PIR Motion Sensor
- Potentiometer (used as Gas Sensor simulation)
- Green LED
- Red LED
- 220Ω Resistors
- Piezo Buzzer
- Push Button
- 16 x 2 I2C LCD Display
- Jumper wires
- Wokwi simulator

Circuit Diagram:



<i>Component</i>	<i>ESP32 Pin</i>
<i>PIR OUT</i>	<i>GPIO 27</i>
<i>Potentiometer SIG</i>	<i>GPIO 34</i>
<i>Green LED</i>	<i>GPIO 25</i>
<i>Red LED</i>	<i>GPIO 26</i>
<i>Buzzer</i>	<i>GPIO 14</i>
<i>Reset Button</i>	<i>GPIO 33</i>
<i>LCD SDA</i>	<i>GPIO 21</i>
<i>LCD SCL</i>	<i>GPIO 22</i>

Working Principle:

- PIR sensor detects motion using hardware interrupts
- Motion detection turns ON green LED (Smart light)
- Potentiometer simulates gas/smoke sensor

- If gas value crosses threshold, emergency override activates
- Green LED turns OFF immediately
- Red LED blinks and buzzer sounds
- LCD shows system status
- Reset button restores safe mode

Code:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

#define PIR_PIN 27
#define GAS_PIN 34
#define GREEN_LED 25
#define RED_LED 26
#define BUZZER 14
#define RESET_BUTTON 33

LiquidCrystal_I2C lcd(0x27, 16, 2);

volatile bool motionDetected = false;
bool emergencyState = false;

void IRAM_ATTR pirISR() {
    motionDetected = true;
}

void setup() {
    Serial.begin(115200);

    pinMode(PIR_PIN, INPUT);
    pinMode(GAS_PIN, INPUT);
    pinMode(GREEN_LED, OUTPUT);
    pinMode(RED_LED, OUTPUT);
    pinMode(BUZZER, OUTPUT);
    pinMode(RESET_BUTTON, INPUT_PULLUP);

    attachInterrupt(digitalPinToInterrupt(PIR_PIN), pirISR, RISING);

    lcd.init();
    lcd.backlight();

    lcd.setCursor(0, 0);
    lcd.print("System Booting");
    lcd.setCursor(0, 1);
    lcd.print("Please Wait");
    delay(2000);

    lcd.clear();
```

```

}

void loop() {
  int gasValue = analogRead(GAS_PIN);

  Serial.print("Gas Value: ");
  Serial.println(gasValue);

  // Reset emergency mode
  if (digitalRead(RESET_BUTTON) == LOW) {
    emergencyState = false;

    digitalWrite(RED_LED, LOW);
    digitalWrite(BUZZER, LOW);

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("System Reset");
    lcd.setCursor(0, 1);
    lcd.print("SAFE MODE");
    delay(2000);
    lcd.clear();
  }

  // Emergency Override
  if (gasValue > 2000) {
    emergencyState = true;
  }

  if (emergencyState) {
    digitalWrite(GREEN_LED, LOW);

    digitalWrite(RED_LED, HIGH);
    tone(BUZZER, 1000);

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("GAS ALERT!!");
    lcd.setCursor(0, 1);
    lcd.print("LOCKDOWN");

    delay(500);

    digitalWrite(RED_LED, LOW);
    noTone(BUZZER);

    delay(500);
    return;
  }
}

```

```

}

// Safe Mode
lcd.setCursor(0, 0);
lcd.print("SAFE MODE  ");
lcd.setCursor(0, 1);
lcd.print("Monitoring...");

// Motion Detection
if (motionDetected) {
    digitalWrite(GREEN_LED, HIGH);

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Motion Found");
    lcd.setCursor(0, 1);
    lcd.print("Light ON");

    Serial.println("Motion Detected");

    delay(3000);

    digitalWrite(GREEN_LED, LOW);
    motionDetected = false;

    lcd.clear();
}

delay(200);
}

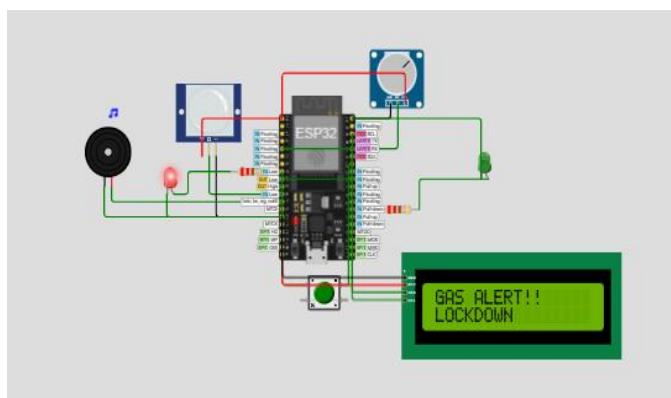
```

Output:

```

Gas Value: 2518
Gas Value: 2730
Gas Value: 2730
Gas Value: 2730
Gas Value: 2730
Gas Value: 2730

```



Result:

The edge – computing smart home appliance was successfully implemented and tested in Wokwi simulation. The system responded instantly to motion and safety conditions.

Conclusion:

Through this project, I learned hardware interrupts, multi – sensor integration, fail – safe system design, and real – time edge computing using ESP32.